Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	*(a)*	(i)		can't be magnesium since gives colour in flame test (1) can't be (strontium) or barium since compound A / the hydroxide is not soluble enough (1)			2	2		2
		(ii)		calcium carbonate accept $CaCO_3$		1		1		1
		(iii)		n(acid) = 3.92×10^{-3} n(compound **C**) = 1.96×10^{-3} (1) M_r(compound **C**) = 56.1 A_r(O) is 16 therefore A_r of metal must be 40.1 (1)			2	2	2	2
	(b)			number of protons = 50 / relative masses of isotopes are 120 and 122 (1) $A_r = \frac{(120 \times 57.9) + (122 \times 42.1)}{100}$ (1) A_r = 120.8 (1) **must** be given to 4 sig figs		3		3	1	